

Mixed chemical bench preparation

Research Lab

REF	ASSESSOR	ASSESSED	NEXT REVIEW
CAT-PW-001	David Kemp	2026-05-30	2027-05-30

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01 OVERVIEW

Business Unit	Research Lab
Location	Chemistry prep room
Risk Assessor	David Kemp
SOP Ref number(s)	SOP-CHEM-015
Date of Assessment	2026-05-30
Date of Review	2027-05-30
Persons at Risk	Staff, Students

Activity Outline

Small-scale preparation, transfer, storage and cleanup of mixed laboratory chemicals for bench work.

02 TASK HAZARDS

Hazard 1 — Glassware breakage and manual handling during chemical transfer

How harm occurs	Cuts or dropped containers could cause injury and secondary chemical release.
Risk (before controls)	9 (L3 × S3)
Controls in place	Use compatible trays, small containers, clear bench space and trained staff only.
Residual risk	2 (L1 × S2)
Further action	—
Owner	—
Due	—
Completed	—

03 COSHH PROCESS STEPS & CHEMICALS

Step 1 — Prepare and transfer mixed chemicals

Engineering controls	Fume hood
PPE	Gloves, Goggles, Lab coat, Face shield
Other step controls	Use compatible secondary containment. Do not open incompatible chemicals together. Confirm glove material and any RPE need against SDS before work.

1.1 Acetone (CAS 67-64-1)

Form / quantity	liquid · 250 mL
H-codes	H225 ; H319 ; H336
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



1.2 Ethanol (CAS 64-17-5)

Form / quantity	liquid · 100 mL
H-codes	H225 ; H319
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



1.3 Dichloromethane (CAS 75-09-2)

Form / quantity	liquid · 50 mL
H-codes	H315 ; H319 ; H335 ; H351
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



1.4 Hydrochloric acid (CAS 7647-01-0)

Form / quantity	liquid · 100 mL
H-codes	H290 ; H314 ; H335
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



1.5 Sulfuric acid (CAS 7664-93-9)

Form / quantity	liquid · 100 mL
H-codes	H290 ; H314
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



1.6 Sodium hydroxide (CAS 1310-73-2)

Form / quantity	solid · 100 g
H-codes	H290 ; H314
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



1.7 Nitric acid (CAS 7697-37-2)

Form / quantity	liquid · 25 mL
H-codes	H272 ; H290 ; H314
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



1.8 Hydrogen peroxide 30% (CAS 7722-84-1)

Form / quantity	liquid · 50 mL
H-codes	H272 ; H302 ; H318
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



1.9 Sodium cyanide (CAS 143-33-9)

Form / quantity	solid · 10 g
H-codes	H300 ; H310 ; H330 ; H410
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



1.10 Phenol (CAS 108-95-2)

Form / quantity	solid · 25 g
H-codes	H301 ; H311 ; H314 ; H331 ; H341 ; H373
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



1.11 Toluene (CAS 108-88-3)

Form / quantity	liquid · 100 mL
H-codes	H225 ; H304 ; H315 ; H336 ; H361 ; H373
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



1.12 Formaldehyde solution (CAS 50-00-0)

Form / quantity	liquid · 25 mL
H-codes	H301 ; H311 ; H314 ; H317 ; H331 ; H341 ; H350
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



1.13 Potassium permanganate (CAS 7722-64-7)

Form / quantity	solid · 20 g
H-codes	H272 ; H302 ; H410
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



1.14 Sodium borohydride (CAS 16940-66-2)

Form / quantity	solid · 10 g
H-codes	H260 ; H301 ; H314
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



1.15 Acetonitrile (CAS 75-05-8)

Form / quantity	liquid · 100 mL
H-codes	H225 ; H302 ; H312 ; H319 ; H332
WEL TWA / STEL	Check EH40/SDS / — (Manual-EH40)
Exposure	Up to 30 minutes per use, Weekly · routes: inhalation, skin, eye

GHS pictograms



04 COSHH ESSENTIALS SCREENING

COSHH Essentials is the HSE control-banding screening tool published in HSG193. It selects a recommended control approach for each substance from the combination of (a) the chemical hazard group derived from CLP H-statements, (b) the scale at which the substance is used, and (c) the volatility band for liquids or the dustiness band for solids. The highest approach across all substances drives the recommended controls.

It is a screening tool only. A competent assessor must verify the recommendation and may impose stricter controls. The scheme does not cover asbestos, lead, pesticides, radioactive materials or biological agents, and does not address physical hazards such as fire, explosion or reactivity.

Recommended approach — Approach 4 — Specialist advice required

Driven by Phenol — hazard group E (H341), scale small, dustiness medium, exposure predictor EP1.

Reference: HSE Generic Guidance Sheet G400 — refer to occupational hygienist / specialist.

Per-substance breakdown

Substance	Group	H-codes	Scale	Band	EP	Approach
Acetone	A	—	small	high (volatility)	EP2	1
Ethanol	A	—	small	high (volatility)	EP2	1
Dichloromethane	D	H351	small	high (volatility)	EP2	3
Hydrochloric acid	C	H314, H335	small	medium (volatility)	EP2	2
Sulfuric acid	C	H314	small	low (volatility)	EP1	1
Sodium hydroxide	C	H314	small	medium (dustiness)	EP1	1
Nitric acid	C	H314	small	medium (volatility)	EP2	2
Hydrogen peroxide 30%	C	H318	small	low (volatility)	EP1	1
Sodium cyanide	D	H300, H310, H330	small	high (dustiness)	EP2	3
Phenol drives controls	E	H341	small	medium (dustiness)	EP1	4
Toluene	D	H361	small	medium (volatility)	EP2	3
Formaldehyde solution drives controls	E	H341, H350	small	high (volatility)	EP2	4
Potassium permanganate	B	H302	small	medium (dustiness)	EP1	1
Sodium borohydride	C	H301, H314	small	medium (dustiness)	EP1	1
Acetonitrile	B	H302, H312, H332	small	high (volatility)	EP2	1

* assumed value used because input was missing or unparseable.

Caveats & assumptions

- Group E hazard (H341) — COSHH Essentials directs to specialist advice (Approach 4). Do not rely on the banded screening.
- H304 aspiration hazard: COSHH Essentials does not fully cover aspiration risk — consider design controls against accidental swallowing/inhalation of mists.
- Group E hazard (H341, H350) — COSHH Essentials directs to specialist advice (Approach 4). Do not rely on the banded

screening.

· COSHH Essentials is a screening tool. A competent risk assessor must verify the recommendation, check the SDS, and may impose stricter controls. Not valid for asbestos, lead, pesticides, radioactive materials, or biological agents.

Reference legend

Hazard group	
Group A	Lower health hazard band: H304, H315, H319, H336 and similar lower-toxicity effects.
Group B	Harmful / STOT category 2 band: H302, H312, H332, H371.
Group C	Toxic, corrosive, serious eye damage, sensitising skin, respiratory irritation or STOT category 1/2: H301, H311, H314, H317, H318, H331, H335, H370, H373.
Group D	Fatal acute toxicity, suspected carcinogen / reproductive toxicant, lactation hazard or repeated-exposure organ damage: H300, H310, H330, H351, H360, H361, H362, H372.
Group E	Respiratory sensitiser, mutagen or carcinogen category 1 / suspected mutagen: H334, H340, H341, H350. Specialist advice is required (Approach 4).

Exposure predictor (EP) band	
EP1	Lowest exposure potential: small amounts with low/medium dustiness, or low-volatility millilitre-scale liquids.
EP2	Low-to-moderate exposure potential: grams of high-dust solids, kg/tonne of low-dust solids, or millilitres of medium/high-volatility liquids.
EP3	Moderate-to-high exposure potential: litre-scale liquid work or kilogram-scale medium/high-dust solid work.
EP4	Highest exposure potential: cubic-metre-scale high-volatility liquids or tonne-scale medium/high-dust solids.

Control approach	
Approach 1	General ventilation and good working practice.
Approach 2	Engineering control, normally local exhaust ventilation or equivalent capture / control.
Approach 3	Containment or enclosure where small breaches may occur.
Approach 4	Specialist advice required: the banding screen alone is not sufficient.

GHS pictogram legend

								
Explosive	Flammable	Oxidising	Gas under pressure	Corrosive	Acute toxicity	Harmful / irritant	Health hazard	Hazardous to environment

05 CONTROL MEASURES

Elimination	Each chemical was reviewed. Remove any substance not required by the SOP before work starts.
Substitution	Substitution must be confirmed against the method and SDS. Use lower hazard alternatives where validated and approved.
Reduction	Use only the minimum working quantity at the bench; return stock containers immediately after dispensing.

Administrative	Only trained and authorised personnel may perform the task. Current SDS for all 15 chemicals must be checked before work; reassess before changing chemicals, quantity, concentration or process.
Air Monitoring	Review EH40/WEL information and SDS exposure limits. Air monitoring is required where exposure cannot be demonstrated as adequately controlled by containment/LEV or after process/control changes.
Health Surveillance	Refer to Occupational Health where SDS, local procedure, sensitiser/CMR classification or potential exposure indicates health surveillance or medical surveillance is required.

06 ADDITIONAL REQUIREMENTS

ChemInventory logged	Yes
SDS version / date	Current supplier SDS to be verified before work · 2026-05-30
Storage	Storage groups generated by CAT must be checked against SDS sections 7 and 10 before use. Keep incompatible groups segregated and retain minimum stock.
Incompatible substances	Segregate acids/bases, flammables/oxidisers, cyanides/acids, water-reactives/water or aqueous wastes, and toxics from incompatible groups unless SDS confirms compatibility.

07 EMERGENCY RESPONSE

Emergency — Spills	Follow SDS Section 6 and the local spill procedure. Only trained staff may tackle small known spills with suitable PPE, ventilation and compatible spill media. Escalate for vapour, toxic, reactive, oxidising, cyanide, drain, fire or uncertainty concerns.
Emergency — First aid	Use SDS Section 4 as definitive first-aid instruction. Move casualty away if safe, irrigate skin/eyes immediately for corrosive exposure, do not induce vomiting unless medically directed, seek urgent medical advice for toxic/CMR/reactive exposures and send SDS with casualty.
Emergency — Fire	Raise alarm and evacuate. Only trained staff may use extinguishers for very small incipient fires where SDS Section 5 confirms suitable media. Do not use water on water-reactive materials unless SDS/emergency responders confirm. Provide SDS and chemical inventory to responders.
Waste handling	Collect waste in compatible, labelled, closed containers under secondary containment. Segregate incompatible waste streams; do not dispose to drain unless SDS/local procedure explicitly permits. Use approved hazardous-waste route.
Other	Emergency shower, eyewash, spill kit and SDS access point must be confirmed before starting.

08 BRIEFING RECORD

Test user · 2026-05-30